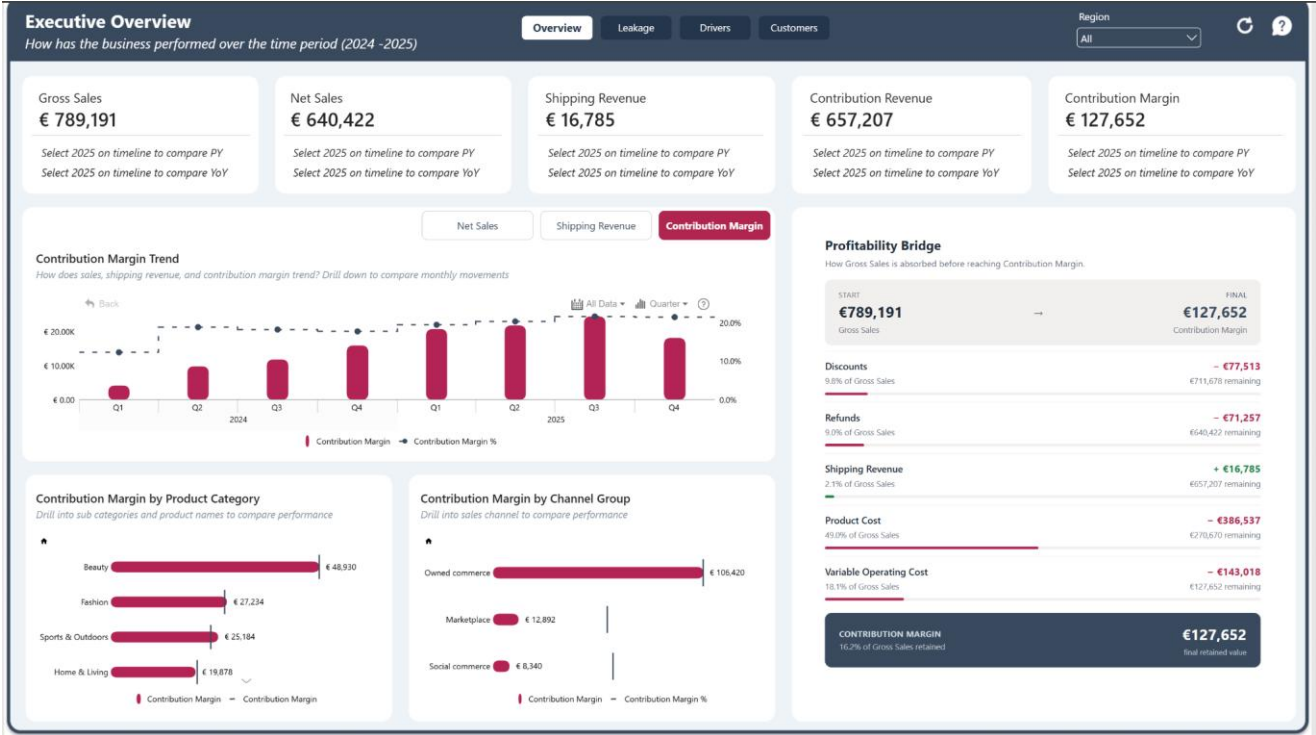


E-Commerce Profitability Analysis

Power BI Portfolio Case Study

Revenue conversion • value leakage • profit drivers • customer profitability

Why isn't the business converting more of its sales into contribution margin, and where can profitability improve?



REPORT
4 pages

PERIOD
2024-2025

TOOLS
Power BI • DAX

DESIGN
ZoomCharts • Star schema

Executive Summary

Turning sales performance into a profitability diagnosis

Project purpose

This project analyses an e-commerce business from Gross Sales through to Contribution Margin to understand why revenue is not translating into stronger profitability. Rather than treating sales performance as the end result, the report separates the problem into four management questions.

Analytical journey

How profitable is the business? → Where is value leaking? → What drives profit? → Are customers creating sustainable value?

What I delivered

- A four-page Power BI report covering Executive Overview, Value Leakage, Profit Drivers and Customer Profitability.
- A star-schema model connecting transactions with products, customers, geography, sales channels, promotions, fulfilment and return reasons.
- Reusable DAX measures covering revenue, contribution margin, returns, lost sales, discounting, fulfilment, acquisition and repeat-customer economics.
- Drill-down hierarchies, interactive navigation, dynamic time controls and contextual KPI reference labels.
- A profitability narrative that separates value creation from value leakage rather than relying on sales totals alone.

Headline results

Gross Sales	€789,191
Net Sales	€640,422
Contribution Revenue	€657,207
Contribution Margin	€127,652
Contribution Margin Rate	19.4%
Product Cost Rate	49.0% of Gross Sales
Discount Cost	€77,513
Return Cost	€102,898
Lost Sales Value	€36,715
Repeat Contribution Margin	€95,028 — 74.4% of total contribution margin

Portfolio summary

The analysis shows that strong sales alone do not guarantee strong profitability. Almost half of Gross Sales is absorbed by product cost, while discounts, returns, lost sales and transaction-related costs create further pressure on margin. Profitability also varies materially by product mix, sales channel and geographic market, while repeat purchasing is a major contributor to customer economics.

1. Business Context and Objectives

From revenue reporting to profitability diagnosis

The dataset captures the commercial journey from gross sales through discounts, fulfilment, refunds, product cost and operating costs to final contribution margin. The analytical challenge was therefore not simply to report what the business sold, but to understand how much value remained after the costs and losses associated with generating and fulfilling those sales.

North-star question

Why isn't the business converting more of its sales into contribution margin, and where can profitability improve?

Primary analytical objective

Determine why the business retains only 19.4% of contribution revenue as contribution margin, identify the largest sources of value erosion, and establish which commercial and customer behaviours are associated with stronger profitability.

Analytical framework

Report page	Management question	Analytical focus
Executive Overview	How effectively are sales converting into margin?	Revenue flow, trend, product and channel sales mix
Value Leakage	Where is value being lost?	Returns, lost sales, discounts, fees and fulfilment
Profit Drivers	Where are stronger profit economics being created?	Products, categories, channels and regions
Customer Profitability	Is the customer base creating sustainable value?	Acquisition, repeat purchasing, age groups and profit at risk

Success criteria

- Follow the financial path from Gross Sales to Contribution Margin.
- Distinguish revenue scale from profitability and quantify the main sources of value leakage.
- Identify products, channels and markets associated with stronger margin performance.
- Connect acquisition and repeat purchasing to customer profitability.
- Allow users to move from executive-level results into lower-level product, city, campaign and customer detail without overcrowding the report.

2. Dataset, Preparation and Semantic Model

Connecting the full path from transaction to contribution margin

The dataset links e-commerce transactions to the commercial and operational factors that influence profitability, including products, customers, markets, promotions, sales channels, fulfilment and returns.

Core analytical areas

Area	Examples
Transactions	Orders, fulfilled units, selling price, gross sales and contribution measures
Products	Brand, brand tier, category, subcategory, product, list price and standard unit cost
Customers	Customer segment, acquisition channel, cohort and age band
Geography	Region, country, city and market attributes
Promotions	Campaign, discount type, discount rate and promotion objective
Sales channels	Channel group, sales channel, device type and payment fees
Fulfilment	Fulfilment model, carrier, service level and cost-to-serve
Returns	Return reason, return group, return loss and recovery attributes
Time / cohort	Calendar periods, cohort age and months since first purchase

Semantic model

The report uses a star-schema design centred on Fact Order Lines, supported by Product, Customer, Geography, Sales Channel, Promotion, Fulfilment, Return Reason, Date and Cohort Age dimensions. Separate helper tables hold core measures, time-intelligence measures, dynamic measures, trend parameters and waterfall steps.

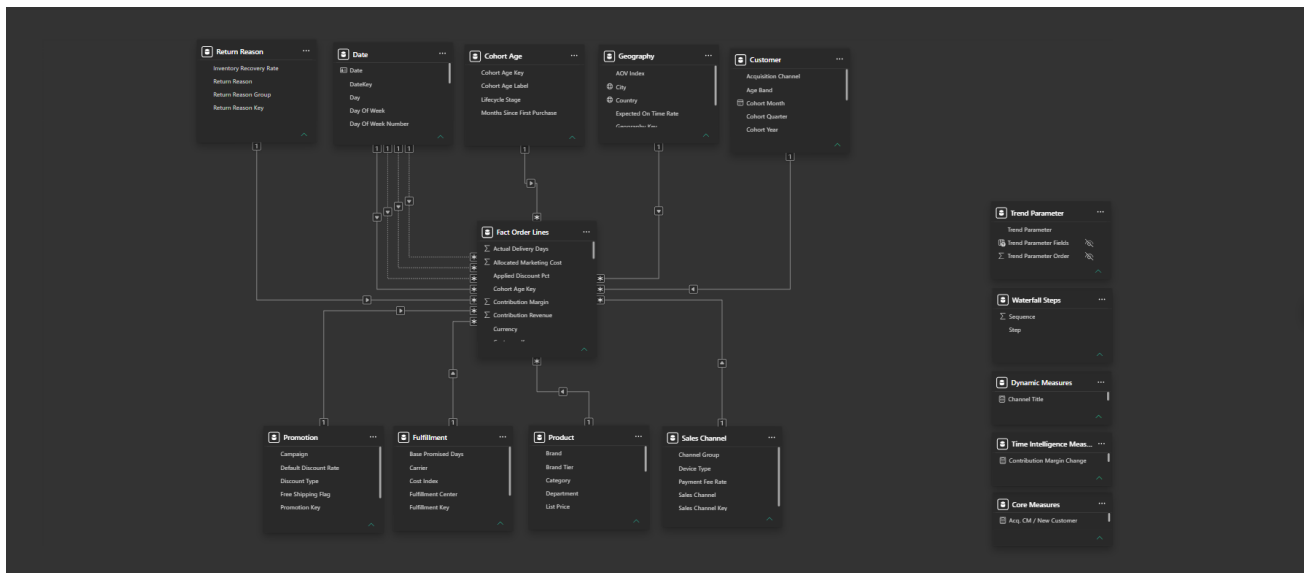


Figure 1. Power BI semantic model

Key modelling considerations

- Contribution Margin is calculated from underlying financial components rather than averaging row-level margin percentages.
- Customer profitability distinguishes acquisition-period value from contribution generated after acquisition.
- Geography used for customer analysis is aligned to the customer's home geography where required.
- Time intelligence supports period comparison while Cohort Age supports repeat-purchase and lifecycle analysis.

3. Measure Design and Analytical Logic

Defining profitability before analysing its drivers

The report follows value from the original sale to the amount retained as contribution margin. This separates revenue leakage from the variable costs required to serve that revenue.

Profitability chain

Gross Sales – Discounts – Refunds + Shipping Revenue = Contribution Revenue → less Product Cost and Variable Operating Cost = Contribution Margin

Core measures

Measure	Purpose
Gross Sales	Original sales value before discounts and refunds
Net Sales	Sales retained after commercial deductions
Contribution Revenue	Revenue available to cover product and operating costs
Contribution Margin	Value remaining after variable costs
Contribution Margin %	Margin efficiency relative to contribution revenue
Return Loss	Financial impact associated with returned orders
Lost Sales Value	Revenue opportunity lost through unfulfilled demand
Acquisition Cost / New Customer	Cost required to acquire each new customer
Repeat Contribution Margin	Margin generated after the customer's acquisition period

Selected DAX patterns

Contribution Margin %

Contribution Margin % =
DIVIDE ([Contribution Margin], [Contribution Revenue])

Repeat Contribution Margin

Repeat Contribution Margin =
CALCULATE ([Contribution Margin], 'Cohort Age'[Months Since First Purchase] > 0)

Company CM% Baseline

Company CM% Baseline =
CALCULATE ([Contribution Margin %], ALL (Geography))

Interaction design

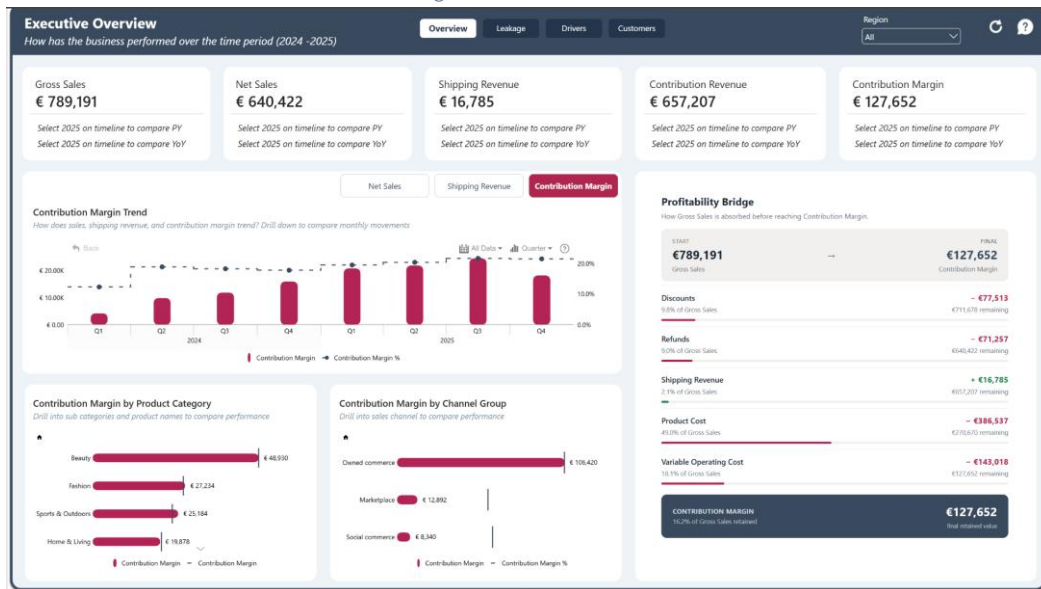
- Year controls support current-period and prior-year comparison.
- Product visuals drill Brand → Category → Subcategory → Product; geography drills Region → City.
- Tooltips provide supporting measures without overcrowding the main chart.
- Page navigation follows Overview → Value Leakage → Profit Drivers → Customer Profitability.

4. Executive Overview and Value Leakage

From overall performance to the specific sources of margin erosion

Executive Overview

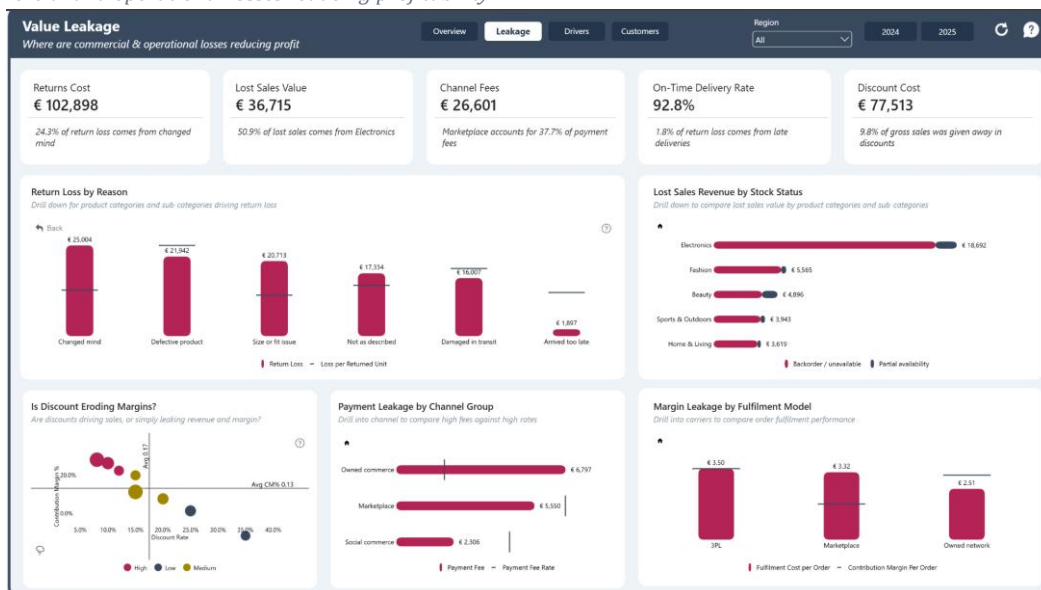
How effectively does sales value convert into contribution margin?



- €789.2K Gross Sales produces €127.7K Contribution Margin; 16.2% of Gross Sales is retained.
- Product Cost is the largest burden at €386.5K, equal to 49.0% of Gross Sales.
- Discounts remove €77.5K and refunds a further €71.3K before final margin.
- Variable Operating Cost absorbs €143.0K, or 18.1% of Gross Sales.

Value Leakage

Where are commercial and operational losses reducing profitability?



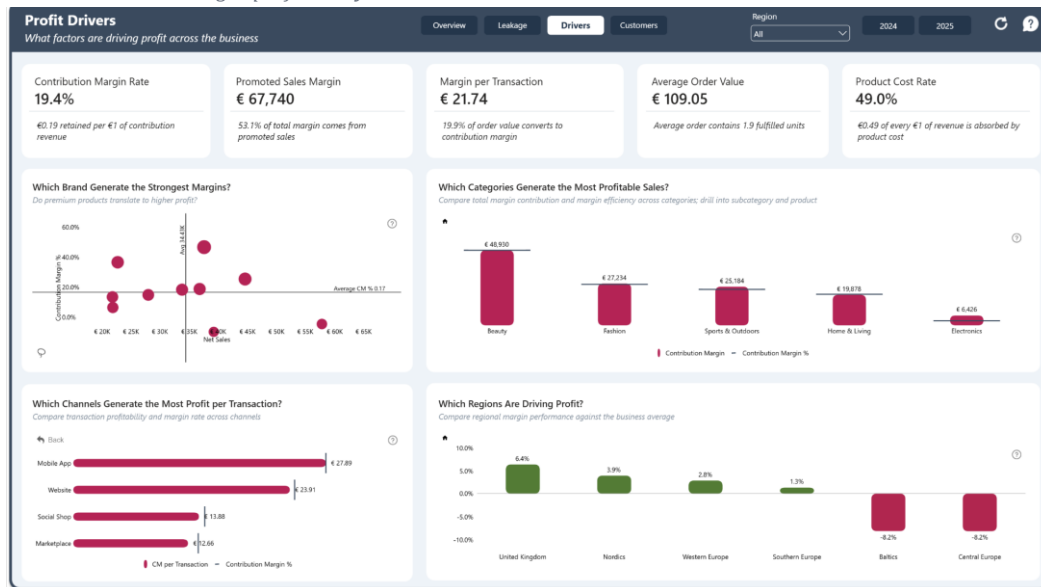
- Returns cost €102.9K; Changed Mind is the largest individual return-loss reason at €25.0K.
- Lost Sales Value totals €36.7K, with Electronics responsible for 50.9% of the total.
- Discount Cost is €77.5K, equal to 9.8% of Gross Sales.
- Channel fees total €26.6K; Marketplace accounts for 37.7% of payment fees.

5. Profit Drivers and Customer Profitability

Where stronger economics are created and whether customer value is sustainable

Profit Drivers

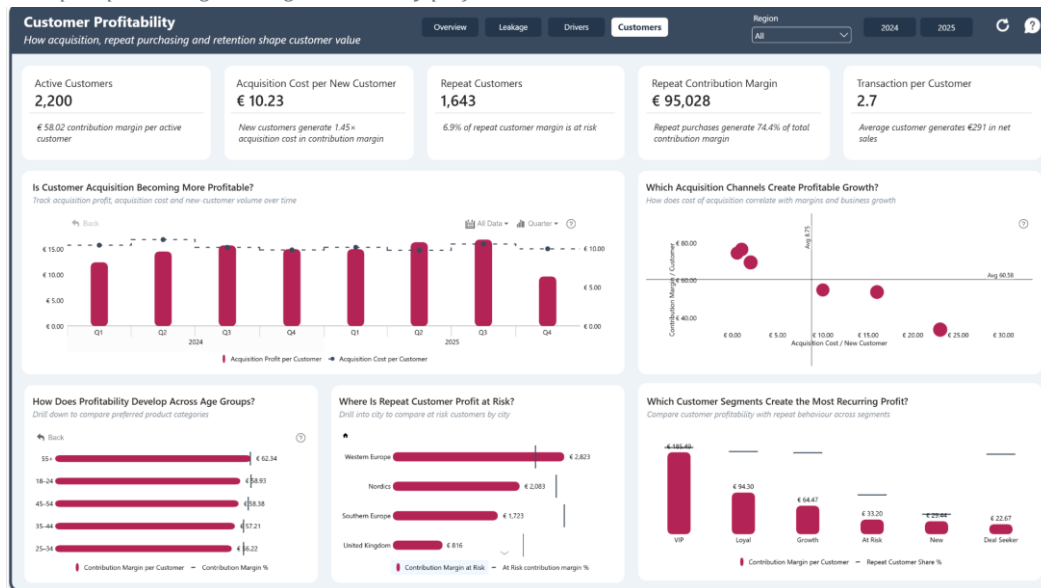
What factors are associated with stronger profitability across the business?



- Beauty contributes €48.9K of margin, ahead of Fashion (€27.2K) and Sports & Outdoors (€25.2K).
- Electronics contributes only €6.4K, highlighting the difference between sales scale and profitability.
- Mobile App has the strongest transaction economics at €27.89 CM per transaction.
- UK margin performance is +6.4pp above the company benchmark; Baltics and Central Europe are about -8.2pp below.

Customer Profitability

Are acquisition and repeat purchasing creating a sustainably profitable customer base?



- 2,200 active customers generate €58.02 Contribution Margin per active customer.
- Repeat purchases generate €95.0K, representing 74.4% of total Contribution Margin.
- New customers generate about 1.45x acquisition cost in Contribution Margin; average CAC is €10.23.
- 6.9% of repeat-customer margin is at risk; VIP customers generate the highest margin per customer.

6. Recommendations

Evidence-led priorities for improving profitability

1. Review product economics, not sales volume alone.

Product Cost absorbs 49.0% of Gross Sales, while some commercially significant categories generate relatively weak Contribution Margin. Electronics, for example, contributes only €6.4K of margin and also accounts for 50.9% of lost sales value. Product and assortment decisions should therefore consider margin quality alongside revenue.

2. Reduce the largest sources of return-related loss.

Returns generate €102.9K in cost, with Changed Mind, Defective Product and Size/Fit issues contributing the largest losses. Drill into category, subcategory and product to identify where product information, quality or fit issues can be addressed.

3. Tighten discounting where margin does not justify the reduction in price.

Discounts remove €77.5K, equivalent to 9.8% of Gross Sales, and the campaign analysis shows weaker margin performance among some of the more heavily discounted promotions. Promotions should be judged on the Contribution Margin they retain, not only the sales they generate.

4. Address inventory-related lost sales, particularly in Electronics.

Lost Sales Value totals €36.7K, with Electronics responsible for 50.9% of the total. Availability planning should prioritise the products and categories where stock constraints are suppressing otherwise valuable demand.

5. Protect repeat-customer profitability.

Repeat purchases generate €95.0K, or 74.4% of total Contribution Margin, while 6.9% of repeat-customer margin is at risk. Retention activity should focus on high-value repeat customers and segments where the financial exposure is greatest.

6. Investigate weaker channel and regional economics before scaling growth.

Mobile App generates €27.89 Contribution Margin per transaction, while regional margin performance ranges from +6.4pp above the company benchmark in the UK to around -8.2pp in the Baltics and Central Europe. Growth decisions should consider both route-to-market economics and regional margin performance.

Decision value

Profitability improvement is unlikely to come from one intervention. The strongest opportunity is to combine better product economics, lower leakage, disciplined discounting, stronger availability and protection of repeat-customer value.

7. Limitations and Project Summary

Transparent assumptions strengthen the credibility of the analysis

Analytical limitations

- Synthetic challenge data: findings demonstrate analytical methods rather than conclusions about a live business.
- Association, not causation: the report identifies relationships and performance differences but cannot prove that one variable directly causes another.
- Contribution Margin is not net profit: fixed overheads, tax, capital costs and other non-variable expenses are outside the analysis.
- Regional variance is relative to the overall company Contribution Margin rate; it is a benchmark signal rather than a prescribed target.
- Contribution Margin at Risk represents value associated with customers classified as At Risk; it is exposure, not a prediction that the value will definitely be lost.
- Acquisition measures use the available reporting period and do not represent a complete customer lifetime-value model.
- Lost sales, return losses and fulfilment measures rely on the definitions supplied in the challenge dataset and would require production validation.

Future enhancements

- Add forecasting for sales, contribution margin and customer value.
- Extend customer analysis into formal cohort-based lifetime value.
- Introduce product-level profitability thresholds and exception reporting.
- Add scenario analysis for discount, product-cost and fulfilment-cost changes.
- Integrate inventory availability with demand forecasting to quantify preventable lost sales.
- Add automated data-quality monitoring and documented measure definitions for production deployment.

Project links

Interactive Power BI report: [View the live dashboard](#)

GitHub repository: [ecommerce-profitability-analysis](#)

Final reflection

The main analytical challenge was not calculating sales or margin individually, but organising the report around the full journey from revenue generation to retained value. Separating the analysis into Executive Overview, Value Leakage, Profit Drivers and Customer Profitability makes it possible to distinguish what the business sells from what it actually retains, where value is being eroded, and which commercial and customer behaviours support stronger profitability.

Project outcome

A concise four-page decision interface that connects financial performance, value leakage, commercial drivers and customer economics in one profitability narrative.